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More rumination, better cows: Exploring genetic parameters of daily rumination time in Holstein dairy cows around first calving

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ABSTRACT

Several studies have indicated that rumination could be used as a metric to monitor dairy cows' performance, condition, and health status. Along with applications in cow management, there may be scope to use rumination data in breeding programs. However, the genetics of rumination around first calving and the relationships with other key traits have not been studied. Therefore, the main objectives of this study were to estimate genetic parameters for daily rumination time (DRT) in Holstein Friesian cows around first calving and genetic correlations with 2 production traits, 3 reproduction traits, and disease frequency. Between April 2020 and January 2024, DRT records were collected from 3,517 Holstein Friesian heifers wearing accelerometer-based collars. Six DRT traits were derived, corresponding to part-periods of the study, namely precalving from −60 to −1 d relative to calving (d 0 = day of first calving), postcalving from 0 to 60 d, precalving1 from −60 to −31 d, precalving2 from −30 to −1 d, postcalving1 from 0 to 15 d, and postcalving2 from 16 to 60 d. Genetic parameters of DRT were estimated based on a random regression test-day model, and approximate genetic correlations were assessed. Daily heritabilities before and after parturition ranged from 0.06 to 0.62, with higher values during −30 to −1 d and lower values during 0 to 15 d. Average heritabilities of precalving1 to postcalving2, calculated based on daily estimates, ranged from 0.07 ± 0.01 (0 to 15 d) to 0.32 ± 0.15 (−30 to −1 d), and their approximate genetic correlations ranged from 0.57 ± 0.07 (precalving1 and postcalving1) to 0.89 ± 0.08 (precalving1 and precalving2). Low to moderate approximate genetic correlations were found for precalving1 to postcalving2 with the average daily milk yield in the first week and month after first calving (−0.09 to 0.36), first calving ease (−0.52 to 0.15),

interval from first calving to first insemination (−0.02 to −0.15), interval from first to last insemination in primiparous cows (−0.03 to −0.17), and disease frequency during 0 to 60 d (−0.05 to −0.34). Cows with higher DRT, particularly from 0 to 60 d, tended to have greater milk yield, better fertility status, and higher disease resistance. These findings indicate that there is scope to use DRT records around first calving in breeding programs to select better cows. Also, DRT patterns before calving may be used to assess the potential calving difficulties of heifers, which would facilitate farm management.

Key words: daily rumination time, random regression model, genetic correlations, parturition

INTRODUCTION

Rumination is at the heart of energy and nutrient supply, productivity, and well-being of dairy cows (Beauchemin, 1991) and is characterized by repetitions of regurgitation, re-chewing, and re-swallowing, resulting in mechanical breakdown of feed particles. Changes in daily rumination time (**DRT**) may be associated with alterations in production and reproductive status, disruptions in rumen functioning, and events imposing stress upon the cow. Several studies have shown that DRT has positive correlations with milk yield (**MY**) in early lactation (Moretti et al., 2016; Stone et al., 2017; Kaufman et al., 2018) and peak lactation (Peiter et al., 2021). One such stressful event is calving, which causes a notable decline in rumination, swiftly succeeded by a rebound after calving (Calamari et al., 2014; Stevenson et al., 2020). The decline in DRT before calving has been associated with calving difficulties (Mammi et al., 2021) and could be used to predict postpartum recovery time (van Dixhoorn et al., 2018) and risk of metritis (Stangaferro et al., 2016; Cocco et al., 2021). Furthermore, lower DRT during the transition period has been associated with higher incidence of ketosis and perinatal metabolic disorders (Kaufman et al., 2016; Antanaitis et al., 2019), rumen acidosis (DeVries et al., 2009; González et

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al., 2012), mastitis (Soriani et al., 2012), and milk fever (Goff et al., 2020).

Daily rumination time stands out as a reliable and earlier indicator of production, reproduction, and health status around calving compared with other indicators (Paudyal, 2021). Although various studies have focused on phenotyping, limited studies have investigated the genetics of DRT. Initial estimates indicate moderate to high heritability for DRT in lactating dairy cows, ranging from 0.32 to 0.34 in Italian Holstein cows (Moretti et al., 2016, 2018), from 0.14 to 0.44 in Danish Holstein cows, and from 0.43 to 0.45 in Belgian Holstein cows (Byskov et al., 2017; Atashi et al., 2024). Genetic correlations between DRT and MY, milk components, and related DMI traits were found to be low to moderate in lactating dairy cows (Moretti et al., 2016, 2018; Byskov et al., 2017). All of these studies on the genetics of DRT included lactating animals only. However, the genetic relationships between DRT and reproduction and disease traits around calving have not been studied, to our knowledge. Understanding these genetic relationships could contribute to early genetic selection of cows with better performance, due to relatively cheap recording of DRT and high reported heritability estimates. Additionally, using DRT data around calving possibly enables the adjustment of cow management to prevent diseases in later phases of the lactation. The period around first calving is especially critical because cows experience a series of stresses for the first time. Individual responses to these stresses are more pronounced around first calving and may be strongly associated with rumination.

Therefore, the objectives of this study were (1) to estimate the genetic parameters for DRT around first calving and (2) to assess the genetic correlations of DRT before and after first calving with production, reproduction, and disease traits of Holstein Friesian cows from a large-scale commercial dairy farm in China.

MATERIALS AND METHODS

Animals, Farm Management, and Feeds

This study was conducted with data from a commercial dairy farm (37.95°N, 118.87°E, Shandong Province, China) collected between April 2020 and January 2024. The farm had ~6,000 lactating Holstein Friesian cows and ~6,000 replacement animals. The average annual MY per cow was ~13,000 kg, slightly above the average of 12,000 kg for registered intensive commercial dairy farms in China (China Dairy Association, 2025). The average MY on test days was 45 kg/cow. All lactating cows were housed in freestall barns and had ad libitum access to TMR and water. Cows were milked 4 times a day (0600, 1200, 1800, and 2400 h) using rotary milk-

ing systems. Cows generally received TMR immediately after milking, whereas heifers were fed TMR at ~0800 and 1500 h. To meet cows' nutritional requirements from precalving and near-calving to postcalving and milking, adjustments were made to the composition of the TMR 30 d before calving, on the day of calving, and 15 d after calving. Heifers in the precalving stage (60 to 31 d before the first calving) were offered a TMR ration containing 22.0 kg DM roughage (i.e., silage, alfalfa, hay, and straw) and 2.6 kg DM concentrates (i.e., brewers grains, corn, meal, and premix) per day (Supplemental Table S1, see Notes). Heifers within 30 d before the first calving were offered a ration containing 18.5 kg DM roughage and 5.2 kg DM concentrates per day. Fresh cows (within 15 d after first calving) were offered a ration containing 17.0 kg DM roughage and 11.4 kg DM concentrates per day. Lactating cows (after 15 d) were offered a ration containing 22.0 kg DM roughage and 17.9 kg DM concentrates per day. All TMR formulations adhered to National Research Council guidelines (NRC, 2001).

Datasets and Quality Control

Rumination time was measured using 3-axis accelerometer-based wearable collars (HR tags, SCR Engineers Ltd., Netanya, Israel) placed on the left side of the heifers' necks. Validation of the equipment revealed high correlation ($r = 0.93$; $R^2 = 0.87$) with direct observations of rumination (Schirmann et al., 2009). The collars recorded rumination time in 2-h periods. Therefore, DRT per heifer was calculated as the sum of the rumination time in the twelve 2-h periods per day and was expressed in minutes per day. Approximately 2 mo before parturition, collars were mounted on the heifers, followed by transfer to the dairy herd. Rumination activity was monitored until 2 mo after calving, resulting in the collection of a total of 464,834 records from 6,668 Holstein Friesian heifers during April 2020 to January 2024.

Additional information on the heifers included days from first calving (DFC, days), average daily MY during the first week (MYFW, kg/d) and the first month (MYFM, kg/d) of the first lactation, age at first calving (AFC, days), calving ease (CE), interval from first calving to first insemination of primiparous cows (ICF, days), interval from first to last insemination of primiparous cows (IFL, days), and disease frequency (DF) during the first 2 mo after first calving. CE is a class variable from 1 to 2, respectively indicating cows calving without and with assistance (limited assistance, requiring 1 person's help for a short period, up to full assistance, requiring 2 or more persons' help, e.g., cesareans). Disease frequency is a class variable from 0 to 1, respectively indicating disease-free cows and cows with at least 1 incident of a disease or metabolic disorder.

Table 1. Basic description of daily rumination time, production and reproduction traits, and disease frequency in Holstein Friesian heifers

Trait ¹	n ²	Mean	Minimum	Maximum	SD
DRT, min/d					
–60 to –1 d	3,517	511	121	759	76
0 to 60 d	3,517	423	2	736	95
MYFW, kg/d	3,008	23.74	5.72	40.18	5.50
MYFM, kg/d	3,324	33.84	10.27	56.12	7.28
AFC, d	3,517	702	634	873	33
CE, d	3,517	1.30	1	2	0.63
ICF, d	3,205	68	62	77	2
IFL, d	3,213	26	0	294	44
DF	3,517	0.20	0	1	0.49

¹DRT = daily rumination time, min/d; MYFW = average milk yield of first week in the first lactation, kg/d; MYFM = average milk yield of first month in the first lactation, kg/d; AFC = age at first calving, d; CE = calving ease, a class variable from 1 to 2, respectively indicating cows calving without and with assistance (limited assistance, requiring 1 person's help briefly, up to full assistance, requiring 2 persons' help during, e.g., cesareans); ICF = interval from first calving to first insemination in primiparous cows, d; IFL = interval from first to last insemination in primiparous cow, d; DF = disease frequency, a class variable from 0 to 1, respectively indicating disease-free cows and cows with at least 1 incident of a disease or metabolic disorder.

²n = number of cows.

Diseases and metabolic disorders were recorded during visual health assessments on the dairy farm, which were conducted daily by a veterinarian. Recorded diseases and metabolic disorders included subclinical and clinical ketosis, uterine inflammation, retained placenta, displaced abomasum, and mastitis. The pedigrees of heifers in this study included 159,252 females and 2,221 sires.

During data cleaning, the first 3 d of DRT measurements were excluded for each heifer, as this period was dedicated to checking and tuning the collars to ensure device functionality. This period was sufficient for the animals to become accustomed to wearing collars. Only records within 60 d before the first calving until 60 d after calving were considered in the analysis (DFC = –60 to 60 d; d 0 = day of calving). To eliminate potential outliers (low values) caused by power outages, collar drops, and other farm accidents during –60 to 60 d, the DRT records beyond mean $\pm 3 \times$ SD for each day were excluded. Individual heifers were included in the analysis if they had at least 5 d of DRT records for –7 to 0 d and 5 d of DRT records for 0 to 7 d, as well as more than 30 d of DRT records over the full period from –60 to 60 d, to ensure sufficient individual records across time. Furthermore, AFC values ranging from 634 to 873 d were retained in this study. Values of MYFW and MYFM within mean $\pm 3 \times$ SD were kept for analysis.

The final dataset after data cleaning had a total of 308,975 DRT records from 3,517 Holstein heifers, which were collected from July 2020 to January 2024. To better fit the V-shaped pattern reflecting the decline and rebound in DRT around first calving, the data were divided into 2 periods: –60 to –1 d (precalving) and 0 to 60 d (postcalving). The number of records was 138,315 for the precalving period and 170,660 for the postcalving

period, and the number of heifers with available data was 3,517 for both (Table 1). Further analyses were performed by splitting the precalving and postcalving periods into smaller periods to assess genetic correlations of rumination around the calving, namely **precalving1** for –60 to –31 d, **precalving2** for –30 to –1 d, **postcalving1** for 0 to 15 d, and **postcalving2** for 16 to 60 d. These 4 periods corresponded to the different TMR diets (Supplemental Table S1).

Genetic Analyses of Daily Rumination Time

A single-trait random regression model (RRM; referred to as model 1) was used to estimate variance components for the precalving and postcalving periods separately, employing the Bayesian method based on the Gibbs sampler and Markov chain Monte Carlo (MCMC) algorithm with the RJMC module within the DMU software (Madsen et al., 2006). The MCMC algorithm was run for a total of 300,000 cycles, with a burn-in phase of 100,000 cycles, and samples were taken at intervals of 50 cycles. To evaluate the convergence of the chain, the package “coda” (Plummer et al., 2005) was used in the R software (R Foundation for Statistical Computing) according to the criteria specified in Geweke (1991). The DFC (–60 to –1 or 0 to 60 d) was normalized in the range of –1 to 1 and subsequently transformed into different orders of Legendre polynomials. Then Legendre polynomials were used to fit the observed DRT patterns in the precalving and postcalving periods. Zero to fifth orders of Legendre orthogonal polynomial functions (Supplemental Table S2, see Notes) were tested for precalving and for postcalving in model 1, and Akaike information criterion (AIC; Akaike, 1974) and Bayesian information

criterion (BIC; Schwarz, 1978) were assessed to the most appropriate order. Lower AIC and BIC values mean the model fits the DRT pattern better. Model 1 is as follows:

$$y_{ijklmn} = TYM_i + CM_j + Feed_k + \sum_{l=0}^2 \beta_{jl} DFC_l + \sum_{l=0}^2 DFC_l a_{lm} + \sum_{l=0}^2 DFC_l pe_{lm} + \varepsilon_{ijklmn},$$

where y_{ijklmn} is the DRT recording; TYM_i is the fixed effect of the test year-month i (42 levels, July 2020–September 2021, November 2021–January 2024); CM_j is the fixed effect of calving month j (12 levels); $Feed_k$ is the fixed effect of feed k (2 levels precalving: –60 to –31 and –30 to –1 d; and 2 levels postcalving: 0–15 and 16–60 d); DFC_l (days from calving) is the covariate effect of the l th Legendre orthogonal polynomial ($l = 0$ –2 for precalving and 0–2 for postcalving) based on DFC (–60 to –1 or 0–60 d); β_{jl} is the l th fixed regression coefficient specific to the j th calving month; a_{lm} and pe_{lm} are the additive genetic and permanent environmental random regression coefficients of the DFC of –60 to –1 or 0 to 60 d, respectively. Two groups of heterogeneous residual variance were considered in ε_{ijklmn} , corresponding to changes of TMR diets supplied: in precalving analysis –60 to –31 and –30 to –1, and in postcalving analysis 0 to 15 and 16 to 60 d. To verify the estimates achieved using random regression, genetic parameters were further estimated for each individual day using an animal model (referred to as model 2), which involved effects similar to model 1, without fixed effect of DFC and permanent environmental effect. Concerning the random effects, the following prior distributions were assumed: $a \sim N(0, \mathbf{G} \otimes \mathbf{A})$, $pe \sim N(0, \mathbf{P} \otimes \mathbf{I})$, and $\varepsilon \sim N(0, I\sigma_e^2)$, where $\mathbf{G}$ and $\mathbf{P}$ represented the covariance matrices linked to the random regression polynomial coefficients for animal additive genetic and permanent environmental effects, respectively. All other parameters were as previously defined. Additionally, we assumed that $\mathbf{G}$ and $\mathbf{P}$ followed the inverted Wishart (IW) distribution, IW ($\mathbf{v}_G, \mathbf{V}_G$) and IW ($\mathbf{v}_{PE}, \mathbf{V}_{PE}$), where $\mathbf{v}$ is the scale matrix and $\mathbf{V}$ is the degrees of freedom ($\mathbf{v} = 2$). The assumption for σ_e^2 was that it followed a scaled inverse chi-squared distribution.

Following the analysis of model 1, the additive and permanent environmental variances and covariances were calculated for a singular time point based on the posterior mean of the (co)variance components estimated for the random regression coefficients of these effects, specifically shown as $\lambda = \mathbf{TGT}'$ and $\xi = \mathbf{TPT}'$, where λ and ξ denote the matrices for additive genetic and permanent environmental (co)variance within –60 to –1 d or 0 to 60 d, $\mathbf{T}$ is the optimal orthogonal polynomials, and $\mathbf{G}$ and $\mathbf{P}$ were as defined previously (Schaeffer, 2004). Re-

sidual variances were also obtained from the posterior marginal distribution samples. Therefore, the daily heritability and repeatability (Rep) in model 1 could be calculated, as well as the average variance components and genetic parameters for different periods (precalving to postcalving), based on the following formulas: $h^2 = \sigma_a^2 / \sigma_p^2$ and $\text{Rep} = (\sigma_a^2 + \sigma_{pe}^2) / \sigma_p^2$. Furthermore, the EBV of the individual DRT were computed for a singular time point using $\text{EBV} = \mathbf{ZT}'$, where $\mathbf{Z}$ is the vector of the solution of additive genetic random regression coefficients specific to animals from model 1. To facilitate the subsequent estimation of genetic correlations for resilience indicators across lactation stages, the average EBV within the defined periods (precalving1, precalving2, postcalving1, postcalving2) of each individual were obtained.

Approximate Genetic Correlations Between Daily Rumination Time and Other Traits

Approximate genetic correlations with other traits were calculated following the method proposed by Calo et al. (1973) and subsequent studies (e.g., Montaldo et al., 2009; Dikmen et al., 2012; Luo et al., 2021). Such correlations were calculated using the EBV from bulls with a reliability exceeding 0.30 for both traits. The average reliabilities for traits of DRT ranged from 0.50 to 0.80, and those for other trait pairs are shown in Supplemental Table S3 (see Notes). The formula is described as follows:

$$r_{g1,2} = \left(\sqrt{(\Sigma RL_1) \% (\Sigma RL_2) \% r_{1,2}} \right) / \Sigma (RL_1 \% RL_2),$$

where $r_{g1,2}$ is the approximate genetic correlation between traits 1 and 2; RL_1 and RL_2 are the reliabilities of EBV for traits 1 and 2; and $r_{1,2}$ is the Pearson correlation coefficient between EBV for traits 1 and 2. The SE of the approximate genetic correlations were estimated as

$$\text{SE} = \sqrt{(1 - r_{g1,2}) / (n - 2)},$$

where n is the number of individuals with records (Sokal and Rohlf, 1995). A single-trait linear animal model with the AI-REML procedure within the DMU software (Madsen et al., 2006) was used to estimate genetic parameters and EBV for other traits. For reproduction traits, fixed effects in the CE and ICF model included year-month of first calving and AFC, whereas animal models of IFL additionally included fixed effects of year-month of first insemination for cows in their first lactation. For the traits of MYFW, MYFM, and DF, the fixed effects of year-month of first calving and AFC were incorporated

in the model. In all models, the random effects comprised the animal additive genetic and residual effects. The factors considered in these genetic models were based on previous studies regarding genetic estimation of fertility in the Chinese Holstein population (Guo et al., 2014; Liu et al., 2017). Finally, the obtained EBV and reliability for precalving1 to postcalving2 and other traits were used in the calculation of the approximate genetic correlation (r_g) and SE.

RESULTS

Phenotypic Description of Daily Rumination Time

The daily rumination patterns of Holstein Friesian heifers were assessed, and combining the records from 2-h intervals per day yielded individual DRT values that varied from 2 to 759 min/d, with a mean of 462 min/d over -60 to 60 d. The mean DRT was 511 min/d precalving and 423 min/d postcalving. The resulting DRT dataset was considered to follow a normal distribution (Figure 1a). The average DRT for each day ranged from 247 to 533 min/d and appeared to be relatively stable during most of the period before calving. A slight decrease in DRT was observed in the days leading up to calving, followed by a significant reduction on the day of calving, and a rebound thereafter (Figure 1b).

Genetic Parameters for Daily Rumination Time

Considering that the differences in AIC and BIC across second to fifth orders of Legendre orthogonal polynomial functions were only 0.01% to 0.02%, along with model complexity, second-order Legendre orthogonal polynomials were fitted from -60 to -1 d and from 0 to 60 d, relative to calving. Estimated variance components based on RRM are presented in Figure 2. Additive genetic variance rapidly increased from -20 to 0 d, whereas both permanent environmental variance and residual variance ($\sigma_e^2 = 2,864$ to 3,270) showed a stable pattern from -60 to

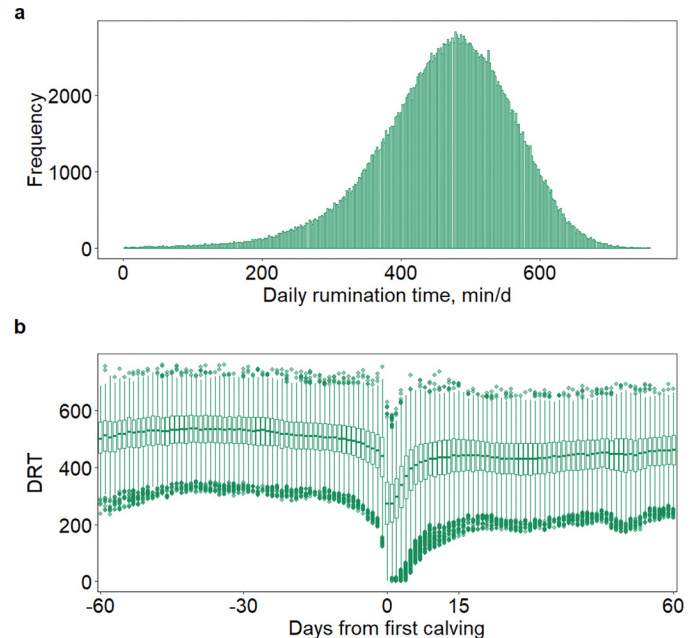


Figure 1. Frequency distribution of daily rumination time (a) and the pattern of daily rumination time (DRT) during -60 to 60 d relative to calving (b) in Holstein Friesian heifers.

-1 d. Following calving, additive genetic variance slowly increased from a low level, whereas both permanent environmental variance and residual variance decreased from 0 to 60 d (Figure 2b). Notably, the highest value in residual variance was reached during 0 to 15 d ($\sigma_e^2 = 11,515$), which was 3 to 4 times higher than precalving levels. Following the decrease in heritability and repeatability during -60 to -31 d, both parameters increased thereafter (-30 to -1 d; Figure 3a). After calving, both heritability (0.06–0.43) and repeatability (0.16–0.54) steadily increased (Figure 3b).

Genetic correlations between different days were calculated based on the days' genetic variances from model 1, which ranged from 0.09 to 0.99 during -60 to -1 d and 0.36 to 0.99 during 0 to 60 d (Supplemental Figure S2).

Table 2. Means and SE of the estimated daily genetic parameters and variance components for daily rumination time during 6 periods around first calving

Period (days relative to first calving)	Variance component ¹			Genetic parameter ²	
	σ_a^2	σ_{pe}^2	σ_e^2	h^2 (SE)	r_e (SE)
-60 to -1 d (precalving)	1,689	924	3,060	0.27 (0.12)	0.44 (0.10)
0 to 60 d (postcalving)	1,588	1,227	5,495	0.23 (0.12)	0.38 (0.11)
-60 to -31 d (precalving1)	1,091	1,099	2,864	0.21 (0.03)	0.43 (0.06)
-30 to -1 d (precalving2)	2,287	749	3,270	0.32 (0.15)	0.45 (0.12)
0 to 15 d (postcalving1)	998	2,105	11,515	0.07 (0.01)	0.21 (0.04)
16 to 60 d (postcalving2)	1,798	916	3,354	0.29 (0.07)	0.44 (0.04)

¹ σ_a^2 = additive genetic variance; σ_{pe}^2 = permanent environmental variance; σ_e^2 = residual variance.

² h^2 = heritability; r_e = repeatability.

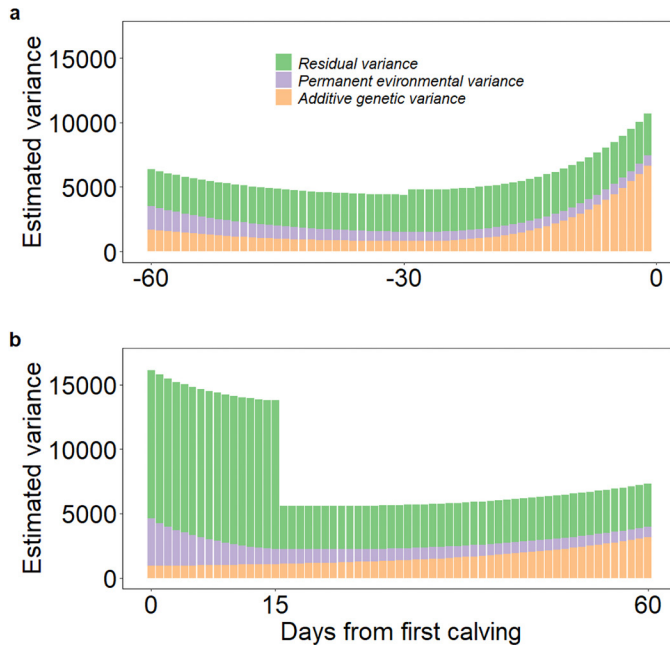


Figure 2. Trends in variance components during -60 to -1 d relative to calving (a) and 0 to 60 d relative to calving (b) for daily rumination time, based on a random regression model in Holstein Friesian heifers.

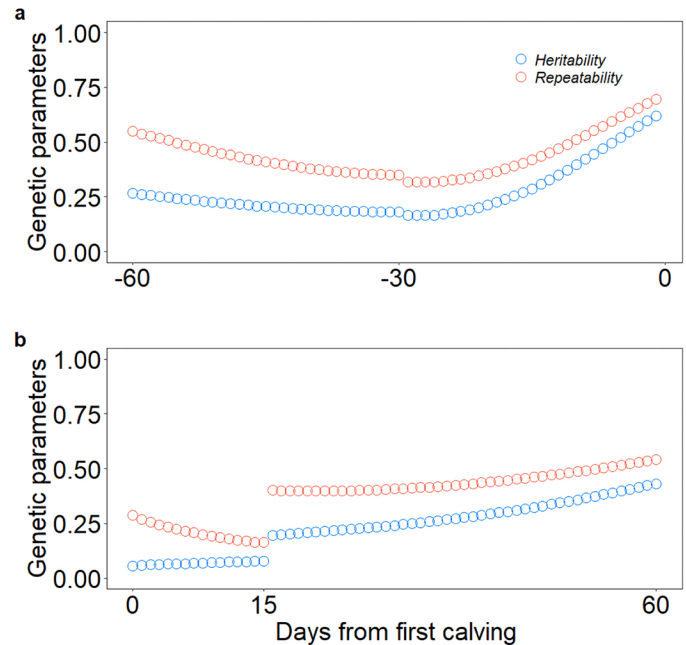


Figure 3. Trends in genetic parameters during -60 to -1 d relative to calving (a) and 0 to 60 d relative to calving (b) for daily rumination time, based on a random regression model in Holstein Friesian heifers.

Correlations were high between adjacent days and decreased when days were further apart, especially during the time before calving. Moderate to high approximate genetic correlations (0.57 ± 0.07 to 0.89 ± 0.08) were found among precalving1, precalving2, postcalving1, and postcalving2 (Table 3).

Approximate Genetic Correlations Between Daily Rumination Time and Other Traits

Approximate genetic correlations were estimated between precalving1, precalving2, postcalving1, and postcalving2 and traits related to production, reproduction, and DF (Table 4). Generally, these genetic correlations were low to moderate and varied across the periods, with lower correlations in -60 to -1 d and higher correlations in 0 to 15 d (postcalving1) and 16 to 60 d (postcalv-

ing2). For the production traits, the genetic correlation between precalving DRT, corresponding to -60 to -1 d, and MYFW and MYFM were low, whereas higher and positive correlations were observed between DRT and MYFW and MYFM after calving (0.16 ± 0.09 to 0.36 ± 0.08). For the reproduction traits, CE generally had associations with DRT, showing a negative correlation (-0.23 ± 0.10) with DRT within 1 mo before calving (precalving2), and a stronger negative relationship (-0.52 ± 0.12) with DRT during 0 to 15 d, compared with other periods (Table 4). Furthermore, ICF and IFL had low and negative genetic correlations with DRT in the different periods, with a range from -0.02 ± 0.13 to -0.15 ± 0.12 (ICF) and from -0.03 ± 0.12 to -0.17 ± 0.12 (IFL). We found that DF had a more negative genetic correlation (-0.21 ± 0.12 to -0.34 ± 0.11) with DRT during 0 to 60 d than during -30 to -1 d (-0.08 ± 0.11).

Table 3. Approximate genetic correlations ($r_g \pm SE$) based on EBV of bulls for daily rumination time among the 4 periods of -60 to -31, -30 to -1, 0 to 15, and 16 to 60 d relative to first calving

Period ¹	-60 to -31 d (precalving1)		-30 to -1 d (precalving2)		0 to 15 d (postcalving1)	
	n ²	r_g (SE)	n	r_g (SE)	n	r_g (SE)
-30 to -1 d (precalving2)	141	0.89 (0.08)	—	—	—	—
0 to 15 d (postcalving1)	128	0.57 (0.07)	131	0.76 (0.06)	—	—
16 to 60 d (postcalving2)	141	0.77 (0.05)	209	0.79 (0.04)	131	0.87 (0.08)

¹Day 0 = day of first calving.

²n = total number of bulls.

Table 4. Approximate genetic correlations ($r_g \pm \text{SE}$) based on EBV of bulls between daily rumination time from 4 periods (–60 to –31, –30 to –1, 0 to 15, and 16 to 60 d relative to first calving) and 2 production traits, 3 reproduction traits, and disease frequency

Trait ¹	Period							
	–60 to –31 d (precalving1)		–30 to –1 d (precalving2)		0 to 15 d (postcalving1)		16 to 60 d (postcalving2)	
	n ²	r_g (SE)	n	r_g (SE)	n	r_g (SE)	n	r_g (SE)
Production								
MYFW	126	–0.09 (0.09)	130	–0.02 (0.09)	127	0.36 (0.08)	130	0.31 (0.08)
MYFM	139	0.07 (0.08)	180	0.01 (0.07)	129	0.16 (0.09)	180	0.29 (0.07)
Reproduction								
CE	155	0.15 (0.13)	155	–0.23 (0.10)	155	–0.52 (0.12)	155	–0.11 (0.14)
ICF	109	–0.02 (0.13)	109	–0.02 (0.11)	109	–0.13 (0.14)	109	–0.15 (0.12)
IFL	106	–0.03 (0.12)	106	–0.04 (0.13)	106	–0.17 (0.12)	106	–0.15 (0.11)
DF	108	–0.05 (0.11)	108	–0.08 (0.11)	108	–0.34 (0.11)	108	–0.21 (0.12)

¹MYFW = average milk yield of first week, kg/d; MYFM = average milk yield of first month, kg/d; CE = calving ease, a class variable from 1 to 2, respectively indicating cows calving without and with assistance (limited assistance, requiring 1 person's help briefly, up to full assistance, requiring help by 2 or more persons during, e.g., caesarians); ICF = interval from first calving to first insemination in primiparous cows, d; IFL = interval from first to last insemination in primiparous cows, d; DF = disease frequency, a class variable from 0 to 1, respectively indicating disease-free cows and cows with at least 1 incident of a disease or metabolic disorder.

²n = total number of bulls.

DISCUSSION

Phenotypic Description of Daily Rumination Time

This study aimed to estimate the genetic parameters for DRT around first calving and the genetic correlations with 2 production traits, 3 reproduction traits, and DF, because first calving involves a series of stressful events the animal has not experienced before. Phenotypes were collected from a single farm, as only very few farms in China have cows equipped with sensor technology that allows DRT recording. The farm under study can be considered representative for large commercial farms in the Northwestern region of China. A future study with multiple farms involved would be desirable and can be performed when more farms are using DRT recording equipment.

A V-shaped DRT pattern was observed during the transition period (–7 to 7 d; Barrier et al., 2013). The mean DRT of 423 min/d postcalving in this study, equivalent to 7 h/d, fell within the range for mean DRT reported in the literature of 2.5 to 8.5 h/d in lactating cows (Adin et al., 2009; Beauchemin, 2018). Rumination activity after first calving is notably reduced compared with the DRT before calving, which might have been strongly connected with the 25% reduction in the average NDF percentage in the TMR. The decreasing trend in average DRT before calving was more pronounced and rapid in the days close to calving (Figure 1b), indicating that calving was imminent. This observation aligns with previous studies reporting a decrease in rumination time as a predictive sign of imminent calving (Schirmann et al., 2013; Borchers et al., 2017; Rutten et al., 2017). A sharp increase in DRT was seen in the first week after calv-

ing, attributed to the increased energy requirements for milk production, basic metabolism, and other activities in fresh cows (Soriani et al., 2013; Sammad et al., 2022). Several studies have linked slow recovery of rumination during early lactation in dairy cows to the occurrence of ketosis and perinatal metabolic disorders in early lactation (Kaufman et al., 2016), uterine diseases (Paudyal et al., 2016), postpartum recovery time (van Dixhoorn et al., 2018), and estrus (Reith and Hoy, 2012; Dolecheck et al., 2015; Pahl et al., 2015). It should be noted that the potential associations between DRT and diseases and reproduction performance after calving may be confounded with other factors, such as negative energy balance and difficult calving.

Genetic Parameters for Daily Rumination Time

This study is the first to investigate the genetic performance of DRT in heifers before and after calving. Both models 1 and 2 yielded similar trends in genetic variances and heritability. Compared with the animal model, the RRM incorporated permanent environmental effects and corrected for trend effects, which improved the reliability of heritability estimates with around 0.1 in this study. Furthermore, applying RRM has the advantage of accounting for genetic relationships among the days of the DRT and obtaining more accurate EBV (Jamrozik et al., 1997). A recent study used RRM to estimate the heritability of weekly average DRT among lactating dairy cows in the Walloon region of Belgium, with a mean of 0.45 in the first parity and 0.43 in the second parity (Atashi et al., 2024), which were similar to heritabilities found around peak lactation ($h^2 = 0.35\text{--}0.43$ from 50 to 60 d postpartum) in this study.

The changes in additive genetic variance, permanent environmental variance, and residual variance around calving (Figure 2), compared with the periods more distant from calving (−60 to −31 and 15 to 60 d), are particularly intriguing. Heifers exhibit increased genetic variation of DRT during −30 to −1 d, suggesting a genetically regulated anticipation to calving. This preparation involves complex neural and humoral regulation (Steinhoff et al., 2019), including hormonal fluctuations (i.e., progesterone, estrogen, and prostaglandins; Reith and Hoy, 2018) and low-grade inflammation (Bertoni et al., 2009), which may play a role in both behaviors such as increased standing, tail raising, and reduced appetite (Huzzey et al., 2005; Wehrend et al., 2006) and metabolic adaptations as the animal approaches calving. The consistent permanent environmental and residual variance observed from −60 to −1 d suggests that environmental effects on preparation for calving are relatively smaller.

The adaptation to the new metabolic state during the first 15 d postpartum was largely affected by environmental effects, as indicated by high estimates of residual and permanent environmental variance. Lactogenesis, uterine involution, and the accompanying changes in endocrine and metabolic states create a unique set of challenges for the fresh cow, which are highly influenced by periparturient management and nutritional conditions around the transition period. Challenges may be further exacerbated by difficult calving and upregulation of the inflammatory response (Bradford et al., 2015), resulting in long-term impairment of rumination recovery and of immunity in dairy cows (Proudfoot et al., 2009), and ultimately affecting their overall health and their later reproductive and lactation performance (Thompson et al., 1983; Fourichon et al., 1999; Wathes et al., 2007). Furthermore, it appeared that most diseases occurred during the first 15 d postpartum in this study, which likely contributed to the large residual variance in this period. Proper calving management is therefore crucial to prevent difficult calving and related diseases.

Approximate Genetic Correlations Between Daily Rumination Time and Other Traits

After calving, fresh cows gradually resume feed intake and rumination to meet energy and nutrient demands for daily maintenance and milk production (Jensen, 2012). Therefore, it can be hypothesized that a higher DRT is genetically associated with higher milk production, better reproductive performance, and greater disease resistance (Moretti et al., 2016, 2018; Byskov et al., 2017; Stevenson et al., 2020). In this study, positive approximate genetic correlations were found between DRT postpartum, MYFW, and MYFM (Table 4). These findings align with other studies (Moretti et al., 2016, 2018; Byskov et al.,

2017). The genetic correlation between precalving DRT and MYFW and MYFM approached zero; therefore, rumination records collected before lactation have limited value for predicting breeding values of milk production. The CE at first calving had negative approximate genetic correlations with DRT traits, particularly from −30 to 15 d. This suggests that heifers with a genetically higher risk of dystocia generally tend to have low EBV for DRT. Additionally, most diseases and metabolic disorders, including subclinical ketosis, uterine inflammation, and retained placenta, occur during the first 15 d postpartum. This seems a likely explanation for a stronger approximate negative genetic correlation between DF and postcalving1 (0 to 15 d) compared with precalving2 and postcalving2 (Table 4). From a management perspective, cows may need additional care if a large DRT reduction in the first 3 d before calving is followed by a slow recovery afterward. Notably, the heritabilities of MYFW, MYFM, CE, ICF, IFL, and DF traits estimated in this study are similar to published results about milk production (~0.25; Li et al., 2020), fertility (0.03–0.10; Guo et al., 2014; Liu et al., 2017), and DF (0–0.03; Wang et al., 2022) among Chinese Holstein cows, indicating that the obtained EBV of those traits were reasonable and applicable for approximate relationship calculations.

A high DRT during postcalving1 and postcalving2 (0–60 d) appeared to be more favorable compared with a high DRT during precalving1 and precalving2 (−60 to −1 d), as it shows stronger approximate genetic associations with higher MYFW and MYFM, shorter ICF and IFL, lower CE score, and lower DF (Table 4). This outcome suggests that heifers with a higher DRT in early lactation usually perform better overall. Therefore, selecting for high DRT during the early phase of the first lactation in breeding may simultaneously improve production, fertility, and health, enhancing cows' ability to recover from calving. Moreover, although the approximate genetic correlation (−0.23) between precalving2 and CE is not strong, it is still possible to indirectly select easy-calving heifers by monitoring DRT before calving, and this also allows farmers to assess the potential calving difficulties of heifers in advance. Precalving DRT has little to no approximate genetic correlations with postpartum production, other reproduction traits, and DF, which may reflect the different patterns of genetic expression between precalving rumination and other traits postcalving. Overall, these results underscore the potential of sensor-based DRT in dairy breeding and management.

CONCLUSIONS

Daily rumination time around first calving is a heritable trait, particularly in the days preceding calving, with high heritability up to 0.62 in the proximity of calv-

ing, followed by a lower but increasing heritability and repeatability after calving. A higher DRT during 0 to 60 d after calving was genetically associated with higher milk yield, better reproductive performance, and lower disease incidence. However, the genetic correlations of DRT before calving were close to zero with postcalving production, reproduction, and disease performance, except for CE. The results of this study provide scope for developing tailored management for individual heifers based on their rumination activity around calving to realize better production, reproduction, and health later. Additionally, this study is the first to offer genetic evidence supporting the association of rumination data around first calving with production, reproduction, and health performance in early lactation. Its results therefore establish a basis for incorporation of DRT into breeding programs.

NOTES

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Nonstandard abbreviations used: AFC = age at first calving; AIC = Akaike information criterion; BIC = Bayesian information criterion; CE = calving ease; DF = disease frequency; DFC = days from first calving; DRT = daily rumination time; ICF = interval from first calving to first insemination of primiparous cows; IFL = interval from first to last insemination of primiparous cows; IW = inverted Wishart; MCMC = Markov chain Monte Carlo; MY = milk yield; MYFM = average daily MY during the first month of first lactation; MYFW = average daily MY

during the first week of first lactation; postcalving1 = period from 0 to 15 d after calving; postcalving2 = period from 16 to 60 d after calving; precalving1 = period from 60 to 31 d before calving; precalving2 = period from 30 to 1 d before calving; r_g = approximate genetic correlation; RRM = random regression model.

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